Low-Light and Infrared Multimodal Remote Sensing in Nighttime Rescue Mission: A Review of Anomaly Detection Methods

Yuting Wan, *Member, IEEE*, Haoyu Yao, Jiawei Liu, Chen Sun, *Graduate Student Member, IEEE*, Ailong Ma, *Senior Member, IEEE*, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— When facing natural disasters such as sudden floods at night, due to sudden natural disasters, high timeliness of rescue information, and complexity and diversity of rescue environment, it is difficult for ground personnel to perform rescue operations in disaster area in time. Remote sensing uncrewed aerial vehicle (UAV) technology plays an escalating role in disaster relief due to fast response and high flexibility advantages. However, with low nighttime visibility level, complex post-disaster environment, and numerous obstructions, traditional UAV's visible light remote sensing struggles to achieve accurate rescue detection at night. Therefore, the multimodal detection methods are investigated using low-light and infrared modalities, exploring the integration of data fusion and detection in night rescue applications, and examining the advantages and disadvantages of different anomaly detection methods here. This article provides the following contributions: 1) a fully-annotated low-light infrared co-observation multimodal remote sensing image dataset for nighttime emergency rescue detection, termed MRSI-NERD; 2) a benchmark test for most state-of-the-art unsupervised anomaly detection methods to thoroughly explore their capability in extracting useful information from normal samples; and 3) a low-light infrared bimodal fusion method based on frequency-domain feature decomposition, which enhances the performance of detectors. The performance of eight types of traditional or deep learning-based detection methods on the MRSI-NERD dataset is reported, including metrics such as receiver operating characteristic (ROC)-area under the curve (AUC), false positive rate (FPR), and true positive rate (TPR). Additionally, a comprehensive analysis of the principles and performance of each category of detection methods is provided.**

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Yuting Wan, Chen Sun, Ailong Ma, and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing and the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China (e-mail: wanyuting@whu.edu.cn; sunchen@whu.edu.cn; maailong007@whu.edu.cn; zhongyanfei@whu.edu.cn).

Haoyu Yao is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, Hubei 430072, China (e-mail: yaohaoyu@whu.edu.cn).

Jiawei Liu is with the School of Cyber Science and Engineering, Wuhan University, Wuhan, Hubei 430072, China (e-mail: 2021302181179@whu.edu.cn).

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I. Introduction

NIGHT emergency rescue scenarios pose great challenges for first responders and rescue teams due to the suddenness of natural disasters [1], the high timeliness of rescue information [2], and the complex diversity of rescue environments [3]. Efficient and accurate rescue operations during the night require advanced technologies that can provide real-time situational awareness, aid in the detection and localization of the disaster targets in low-light and complicated environments [4].

With the continuous development of computer vision, object recognition and detection algorithms based on deep neural networks (DNNs) are being applied in emergency rescue scenarios [5], providing a viable solution to address issues such as the limitation of timeliness in disaster rescue [6]. The role of remote sensing uncrewed aerial vehicle (UAV) technology equipped with deep convolutional neural networks in rescue and disaster relief operations is increasingly significant [7].

In the context of remote sensing and emergency rescue and search scenarios, low-light remote sensing technology and thermal infrared remote sensing technology are two widely applied and mature image recognition methods in the field of traditional image recognition [8]. They exhibit good complementarity and integration [9]. The fusion of low-light and thermal infrared images has become the forefront of the current discipline's development [10]. However, traditional offline remote sensing image recognition methods face challenges in achieving precise detection of rescue targets under extreme conditions such as low nighttime light intensity and complex post-disaster terrain [11].

Low-light and infrared images fusion is a branch of image fusion that has significant applications in remote sensing image recognition [9]. In the context of nighttime emergency rescue, low-light imagery is highly sensitive to lighting conditions [12] but can capture spatial information, texture details more effectively [13]; infrared imagery exhibits prominent targets and is less affected by lighting conditions [14], but it typically has lower image resolution and blurry target backgrounds [15]. The fusion of the above two can effectively combine the

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advantages of both [16], increase the amount of information, and thus improve nighttime rescue operations [17]. The fusion algorithms include traditional fusion algorithms based on spatial or transform domains [18], as well as autoencoder (AE) algorithms [19], convolutional neural network algorithms [20], and adversarial neural network algorithms [21] based on deep learning. Recently, many innovative approaches have emerged in the field of image fusion. For example, the color-aware fusion technique [22] effectively improves the recognizability of targets in low-light environments by preserving the color information and the thermal features. In addition, a hierarchical guided fusion network based on Laplace pyramid [23] enhances the complementarity of cross-modal information while preserving image details through multiscale feature decomposition and hierarchical fusion strategies. Besides, the transformer-based large-scale models for multimodal infrared and visible image fusion have also attracted significant attention. These approaches [24], [25], [26], [27], [28] leverage self-attention mechanisms to effectively integrate complementary features across different modalities. Notably, SwinFusion [24] implements a hierarchical attention framework based on Swin Transformer [25], establishing dual-stream attention modules for cross-domain feature alignment. Furthermore, DATfusion [26] is proposed to preserve important features and global information through the interaction of dual attention with transformer blocks, thereby achieving multimodal image fusion. In addition, recent studies have also proposed a semantic-driven fusion framework [27], which further improves the fusion quality in complex scenes by introducing semantic prior knowledge to guide the feature fusion and provides richer semantic information for target detection. PromptFusion [28] further explores a prompt-based method for guiding image fusion from vision-language models, which aims to enhance target identification and fusion quality by leveraging semantic prompts. Although these fusion methods have achieved remarkable performance in recent decades [29], the existing algorithms still have limitations when it comes to the specific scenario of nighttime emergency rescue [30].

Anomaly detection refers to distinguishing the identification and localization of anomalies in limited or even absence of prior knowledge [31]. Anomaly detection and target detection based on high-resolution remote sensing images can effectively utilize the existing spectral information in the images to accurately locate rescue targets without the need for a large amount of prior information specifically tailored to the targets [32], [33]. This approach better meets the requirements of high timeliness and accuracy in emergency rescue missions [6]. Unsupervised anomaly detection mainly includes methods such as sample reconstruction, pseudo anomaly augmentation, and knowledge distillation [34], [35], [36]. Specifically in nighttime emergency rescue scenarios, the complexity of background information [37], the diversity of anomalous samples [38], and the presence of weak target information pose challenges for the deployment and application of existing anomaly detection algorithms [39].

Considering the aforementioned issues, this article establishes a synchronized low-light and infrared coaxial imaging system and collects a dataset that simulates nighttime emergency rescue scenarios. Furthermore, some advanced unsupervised anomaly detection methods were evaluated on the proposed low-light and infrared datasets. This contributes to the development of a nighttime remote sensing emergency rescue system designed for low visible light conditions. The main contributions of this article are summarized as follows.

1) *A Dataset of Multimodal Infrared and Low-Light Remote Sensing Images Simulating Emergency Rescue Scenarios:* To address the lack of remote sensing image datasets for nighttime emergency rescue scenarios, a highly realistic low-light infrared multimodal nighttime dataset is proposed, termed MRSRI-NERD. It includes various defects such as hypothermia [40], occlusion [41], and fallen rescue targets [42] and is fully annotated to simulate real-world scenarios accurately [44]. To the knowledge, this is currently the only available remote sensing multimodal dataset specifically designed for nighttime emergency rescue scenarios.
2) *A Benchmark for Unsupervised Anomaly Detection Models in Emergency Rescue Datasets:* In this article, a survey of unsupervised image segmentation methods in the existing literature was conducted, identifying challenges and outlining future research directions. In the multimodal remote sensing image dataset for nighttime emergency rescue detection (MRSI-NERD) dataset, four categories of traditional detectors and four categories of deep detectors were comprehensively evaluated, reporting metrics such as area under the curve (AUC) score, false positive rate (FPR), and true positive rate (TPR), to provide a benchmark for extensive research on unsupervised nighttime multimodal detection.
3) *A Dual-Stream Low-Light and Infrared Modality Fusion Network Based on Frequency-Domain Feature Decomposition:* The dual-stream fusion network is predicated on frequency-domain feature decomposition with a Transformer–CNN framework. This enables the extraction and fusion of global and local features, ensuring a more accurate reflection of modality-specific and modality-shared features. Specifically, the work introduces the innovative utilization of invertible residual neural network (IRNN) and Lite-transformer blocks for optimizing the balance between fusion quality and computational cost. The superiority of the proposed fusion method compared to existing algorithms was comprehensively validated through metrics such as entropy (EN), standard deviation (SD), spatial frequency (SF), mutual information (MI), sum of correlation differences (SCDs), and structural similarity index measure (SSIM). Improvements in the AUC metric are reported across multiple downstream anomaly detection algorithms, validating the effectiveness of the approach in the nighttime emergency rescue mission.

II. Construction of Multimodal Dataset for Nighttime Scenarios

Research in the field of night rescue using multimodal low-light and infrared remote sensing has undergone a remarkable evolution. Initially, the focus was primarily on the

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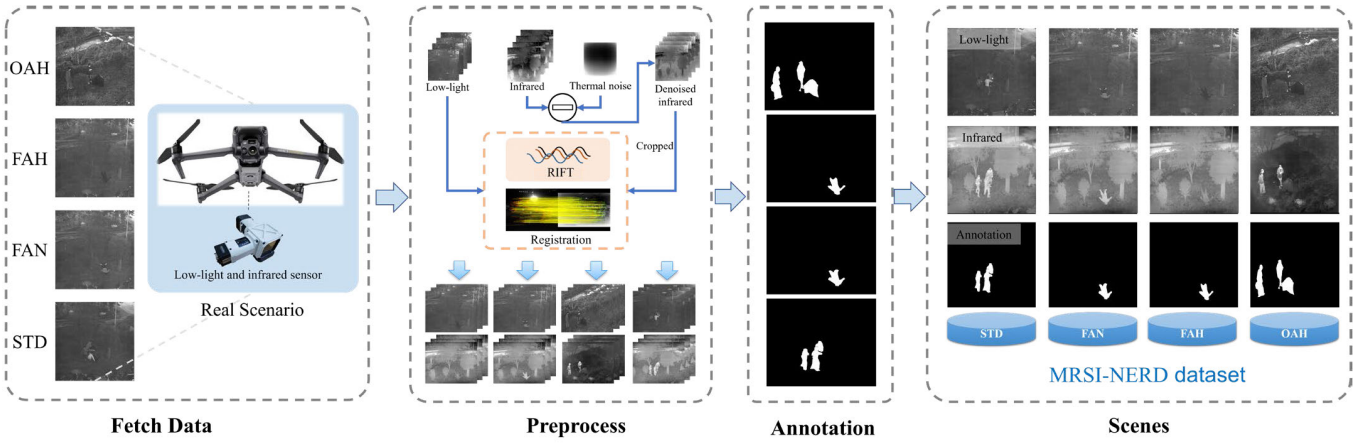


Fig. 1. Construction of the MRSI-NERD dataset. The overall process is comprised of three distinct steps: data fetching, data preprocessing, and annotation. Initially, the original low-light and infrared images were obtained from the device in four different real scenarios. For the preprocessing step, the denoised images were then obtained by subtracting the infrared dark image (solely containing thermal noise) from the original infrared image [51]. Subsequently, the infrared images were subjected to a process of cropping, whereby the dark edges were removed. The RIFT algorithm [52] was then employed to facilitate the alignment of the low-light and infrared images during the registration process. Finally, the synchronized low-light and infrared images, which contained targets, were fully annotated.

stand-alone application of low-light and infrared remote sensing technologies. Low-light remote sensing was explored for its ability to capture spatial and textural detail, although it is highly sensitive to lighting conditions. Infrared remote sensing, on the other hand, was recognized for its ability to detect prominent targets with less dependence on ambient light, although it suffered from issues such as lower resolution and blurred backgrounds.

As the field progressed, the concept of fusing these two modalities emerged. Early attempts involved traditional fusion algorithms based on spatial or transform domains. These aimed to combine the complementary features of low-light and infrared images to enhance the overall information content. However, with the advent of deep learning, more advanced fusion algorithms such as AE, convolutional neural network, and adversarial neural network were proposed. These methods showed promise in improving the quality of fused images and extracting more meaningful features.

Nevertheless, the development of deep and statistical-based image recognition algorithm relies on training on a large amount of valid data [43], but traditional single-modal visible light data are limited by daylight conditions and cannot meet the requirements of nighttime rescue [44]. There is a lack of image data for nighttime emergency rescue scenarios. Thus, in this article, a fully annotated MRSI-NERD was constructed, to address the issue of data scarcity, as shown in Fig. 1. Basically, a total of 4087 frames of video stream data were collected, where 4044 of them are images containing targets, and the remaining 43 frames are background reference images without anomalies, used for training the anomaly detection model. The collected data were divided into four categories of detection tasks: "Fallen And Hypothermic," "Fallen And Normothermic," "Obstructed And Hypothermic," and "Standing," abbreviated as "FAH," "FAN," "OAH," and "STD," respectively.

1) *FAH:* It represents the scenario where the rescue target has fallen and is hypothermic. This situation is common in emergency rescue scenarios, where the target is in an urgent state due to hypothermia [45], but at the same time, it interferes with the detection of infrared detectors, posing a certain level of difficulty. Obtaining images of hypothermic individuals is not a viable option, so this condition was simulated using insulating foil boards, which were cropped into the shape of a person [46].
2) *FAN:* It refers to the scenario where the rescue target has fallen and cannot stand up but still maintains a normal body temperature. In comparison to FAH, the infrared detector can detect the target, but the low-light detector may not effectively capture information about the target [47]. The FAN rescue scenario is designed to simulate real-life conditions [48].
3) *OAH*: Scenario refers to the scenario where the target is obstructed and cannot be detected. In this situation, the infrared detector becomes ineffective due to the thick obstructive material providing thermal insulation. This scenario is commonly encountered in emergency rescue situations, where detecting obstructed rescue targets beyond obstacles is a persistent challenge in the field of emergency rescue. OAH is also widely present in actual emergency rescue scenarios [49].
4) *STD*: It is a more general rescue scenario where the target can still stand and has a normal body temperature, showing a significant contrast with the surrounding environment. In this situation, the detector should exhibit high performance, serving as a criterion to verify the effectiveness of the detector.

A low-light and infrared co-calibrated imaging device [50] was applied to capture data from the aforementioned scenarios, which were processed and annotated appropriately. The data processing workflow is illustrated in Fig. 1. First, raw data were captured from the device. The resolution of low-light images is 2048 × 2048, while the infrared images are 640 × 512. The original infrared images have thermal noise,

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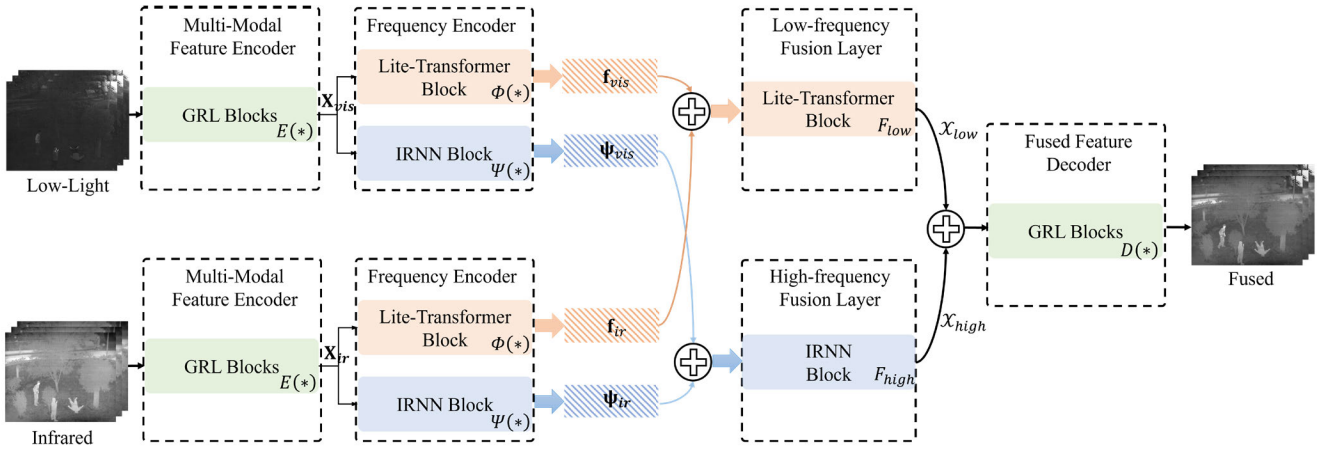


Fig. 2. Architecture of the proposed dual-stream multimodal fusion model based on frequency-domain feature decomposition.

which must be eliminated through a process of denoising. By deactivating the infrared sensor in an environment devoid of light exposure, an infrared dark image could be obtained with solely thermal noise. The denoised images were then obtained by subtracting the infrared dark image (thermal noise) from the original infrared image [51]. Afterward, the resulting infrared images have dark edges that need to be cropped out. Additionally, although the low-light and infrared images are aligned at the center, they have different fields of view, resulting in varying degrees of distortion at the image edges. Therefore, the RIFT algorithm [52] was employed for registration. Finally, the synchronized low-light and infrared images were obtained.

III. Dual-Stream Multimodal Fusion Model

In this section, a dual-stream low-light and infrared modality fusion network is introduced, as shown in Fig. 2, based on frequency-domain feature decomposition, which leverages the complementary information from visible light and infrared modalities to enhance the detection performance [53].

A. Method Overview

The model contains four components:

1) an encoder–decoder module for high-fidelity reconstruction;
2) a low-frequency extractor for extracting global features;
3) a high-frequency extractor for extracting local detailed features;
4) a dual-stream feature fusion module.

The overall workflow is illustrated in Fig. 2, and the network achieves excellent performance on the low-light and infrared fusion task.

B. Multimodal Feature Encoder

For the low-light and infrared inputs, a shared feature backbone encoder was applied to help the subsequent frequency-domain feature extractor for further processing. Initially, the input pair of low-light and infrared images $\mathbf{I}_{\text{vis}}$ and $\mathbf{I}_{\text{ir}}$ was fed into the encoder, respectively, represented as follows:

$$\mathbf{X}_{\text{vis}} = E(\mathbf{I}_{\text{vis}}), \quad \mathbf{X}_{\text{ir}} = E(\mathbf{I}_{\text{ir}}) \tag{1}$$

where $E(*)$ is the encoder and $\mathbf{X}_{\text{vis}}$ and $\mathbf{X}_{\text{ir}}$ are the shared features extracted from low-light images and infrared images, respectively.

For the encoder, the global, regional, and local (GRL) [54] block, a state-of-the-art method adopted in the field of image restoration, is selected. The reason is that GRL can efficiently and explicitly extract high-resolution image features and perform restoration, which is beneficial for improving the quality of image fusion [55].

C. Frequency-Based Feature Extractor

Then, the aforementioned shallow features $\mathbf{X}_{\text{vis}}$ and $\mathbf{X}_{\text{ir}}$ were fed into a low-frequency extractor and a high-frequency extractor, respectively, obtaining the corresponding frequency-domain features, represented as follows:

$$\mathbf{f}_{\text{vis}} = \Phi(\mathbf{X}_{\text{vis}}), \quad \psi_{\text{vis}} = \Psi(\mathbf{X}_{\text{vis}}) \tag{2}$$

$$\mathbf{f}_{\text{ir}} = \Phi(\mathbf{X}_{\text{ir}}), \quad \psi_{\text{ir}} = \Psi(\mathbf{X}_{\text{ir}}) \tag{3}$$

where $\Phi(*)$ represents the low-frequency extractor; $\Psi(*)$ represents the high-frequency extractor; and $\mathbf{f}_{\text{vis}}$ and $\mathbf{f}_{\text{ir}}$ are the extracted low-frequency features of low-light and infrared inputs, respectively. Similarly, ψ_{vis} and ψ_{ir} are the extracted high-frequency features of low-light and infrared inputs, respectively.

A Lite-transformer-based low-frequency extractor was applied because it can improve the speed of inference [56]. The lightweight transformer module allows it to obtain more long-range information [57], corresponding to high-frequency information [58], while its ability to extract short-range information is reduced. This symbolizes a weaker low-frequency information extraction capability. For the high-frequency extractor, an invertible residual neural network [59] was used. The convolutional structure of CNN is effective in extracting short-range information [60], corresponding to high-frequency components [61]. Invertible neural networks have the theoretical capability of lossless information compression [62], enabling the preservation of high-frequency information.

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D. Frequency-Based Fusion Layer

The low-frequency features $\mathbf{f}_{\text{vis}}$ and $\mathbf{f}_{\text{ir}}$ were first concatenated and then fed into the low-frequency fusion layer, while the high-frequency features ψ_{vis} and ψ_{ir} underwent the same operation but were fed into the high-frequency fusion layer. This procession can be described as follows:

$$\begin{aligned}\boldsymbol{\mathcal{X}}_{\text{low}} &= F_{\text{low}}(\text{CAT}(\mathbf{f}_{\text{vis}}, \mathbf{f}_{\text{ir}}))\\ \boldsymbol{\mathcal{X}}_{\text{high}} &= F_{\text{high}}(\text{CAT}(\psi_{\text{vis}}, \psi_{\text{ir}}))\end{aligned} \tag{4}$$

where $F_{\text{low}}(^*)$ is the low-frequency fusion layer, $F_{\text{high}}(^*)$ is the high-frequency fusion layer, $\boldsymbol{\mathcal{X}}_{\text{low}}$ is the fused low-frequency feature, $\boldsymbol{\mathcal{X}}_{\text{high}}$ is the fused high-frequency feature, and CAT$(^*, ^*)$ is the channel concatenation operation.

The low-frequency fusion layer consists of the Lite-transformer-based block, while the high-frequency fusion layer consists of the IRNN block, which aims to fuse low/high frequency features, respectively.

E. Fused Feature Decoder

The fused feature decoder is comprised of the GRL blocks. The frequency-based fused features $\boldsymbol{\mathcal{X}}_{\text{low}}$ and $\boldsymbol{\mathcal{X}}_{\text{high}}$ were concatenated and reduced in dimension, respectively. They were then fed into the fused feature decoder to generate the final fused image, given as follows:

$$\mathbf{I}_{\text{fused}} = D(\text{CAT}(\boldsymbol{\mathcal{X}}_{\text{low}}, \boldsymbol{\mathcal{X}}_{\text{high}})) \tag{5}$$

where $D(^*)$ is the decoder, $\mathbf{I}_{\text{fused}}$ is the final fused image, and CAT$(^*, ^*)$ is the channel concatenation operation.

F. Loss Function

To achieve a high-fidelity output that combines features from both modalities, the network was trained in two stages.

In the first stage, the network's ability was trained to reconstruct images for each modality. This is attained by concatenating high- and low-frequency features and then feeding them into the GRL decoder. The loss function is described as follows:

$$\begin{aligned}L_1 &= \alpha_1 L_{\text{vis}} + \alpha_2 L_{\text{ir}} + \alpha_3 L_{\text{indep}}\\ &\text{s.t. } \alpha_1 + \alpha_2 + \alpha_3 = 1\end{aligned} \tag{6}$$

where L_{vis} and L_{ir} are the reconstruction errors for low-light and infrared images, respectively, and L_{indep} represents the loss that quantifies the independence of low-frequency representations from high-frequency features. α_1–α_3 are the tuning parameters, with the constraint condition of $\alpha_1 + \alpha_2 + \alpha_3 = 1$.

For the reconstruction loss L_{vis}, which is similar to L_{ir} in calculation, it is conducted by mse [64], namely, (7). Among them, $\bar{\mathbf{I}}_{\text{vis}}$ is the reconstructed low-light image and $\mathbf{I}_{\text{vis}}$ is the original image, consistent with the previous definition mentioned above, where μ is the tuning parameter

$$L_{\text{vis}} = ||\bar{\mathbf{I}}_{\text{vis}} - \mathbf{I}_{\text{vis}}||_2^2 + (1-\mu)\text{MSE}(\bar{\mathbf{I}}_{\text{vis}}, \mathbf{I}_{\text{vis}}). \tag{7}$$

For the independence loss, assuming that each pixel of image $\mathbf{I}_{\text{vis}}$ is denoted by x_i and each pixel of infrared image $\mathbf{I}_{\text{ir}}$ is denoted by y_i, the calculation of L_{indep} can be given in (8). It specifies that the independence loss is computed using the cosine distance [63]

$$L_{\text{indep}} = \frac{\sum_i^{x_i y_i}}{\sqrt{\sum_i^{x_i^2}}\sqrt{\sum_i^{y_i^2}}}. \tag{8}$$

In the second stage, the network was trained for multimodal image fusion and feature extraction. The paired low-light and infrared images $\{\mathbf{I}_{\text{vis}}, \mathbf{I}_{\text{ir}}\}$ were fed into a well-trained GRL encoder to obtain the shallow features $\{\mathbf{X}_{\text{vis}}, \mathbf{X}_{\text{ir}}\}$. Then, the low-frequency extractor and the high-frequency extractor were applied to obtain the decoupled features $\{\mathbf{f}_{\text{vis}}, \psi_{\text{vis}}, \mathbf{f}_{\text{ir}}, \psi_{\text{ir}}\}$. Through the fusion layer, $\{\mathbf{f}_{mi}, \mathbf{f}_{\text{ir}}\}$ and $\{\psi_{mi}, \psi_{\text{ir}}\}$ were combined, obtaining the aggregated low-frequency feature $\boldsymbol{\mathcal{X}}_{\text{low}}$ and high-frequency feature $\boldsymbol{\mathcal{X}}_{\text{high}}$, and these combined features were then fed into the GRL decoder to produce the final fused image $\mathbf{I}_{\text{fused}}$.

During this stage, the loss function is defined as follows, which is comprised of intensity loss, gradient loss, and independent loss:

$$L_2 = \lambda_1 L_{\text{VIS}_{\text{int}}} + \lambda_2 L_{\text{IR}_{\text{int}}} + \lambda_3 L_{\text{VIS}_{\text{grad}}} + \lambda_4 L_{\text{IR}_{\text{grad}}} + \lambda_5 L_{\text{indep}}. \tag{9}$$

The intensity loss $L_{\text{VIS}_{\text{int}}}$, which is similar to $L_{\text{IR}_{\text{int}}}$ in calculation, is described as follows:

$$L_{\text{VIS}_{\text{int}}} = \frac{1}{HW}||\mathbf{I}_{\text{fused}} - \mathbf{I}_{\text{vis}}||_2^2 \tag{10}$$

where H and W are the height and width of the image, respectively.

The gradient loss $L_{\text{VIS}_{\text{grad}}}$, which is similar to $L_{\text{IR}_{\text{grad}}}$ in calculation, is described as follows:

$$L_{\text{VIS}_{\text{grad}}} = \frac{1}{HW}||\nabla\mathbf{I}_{\text{fused}} - \nabla\mathbf{I}_{\text{vis}}||_2^2 \tag{11}$$

where ∇ represents the Sobel gradient operator and H and W are the height and width of the image, respectively.

IV. Anomaly Detection Methods in Emergency Rescue Situations

The typical goal of anomaly detection is to identify anomalies that differ from normal samples through unsupervised or semi-supervised methods, performing binary classification or localizing tasks. In recent years, there has been significant progress in image anomaly detection techniques, yet their application in the field of emergency rescue remains unexplored. In this section, unsupervised image anomaly detection methods are reviewed and categorized into traditional methods and deep learning-based methods, distinguishing whether they incorporate DNNs. The overview of the image anomaly detection methods is shown in Fig. 3.

A. Anomaly Detection Methods Based on Traditional Approaches

In this section, traditional image anomaly detection methods are categorized into the following types: statistical model-based, sparse coding reconstruction-based, decomposition-based, and frequency-domain analysis-based.

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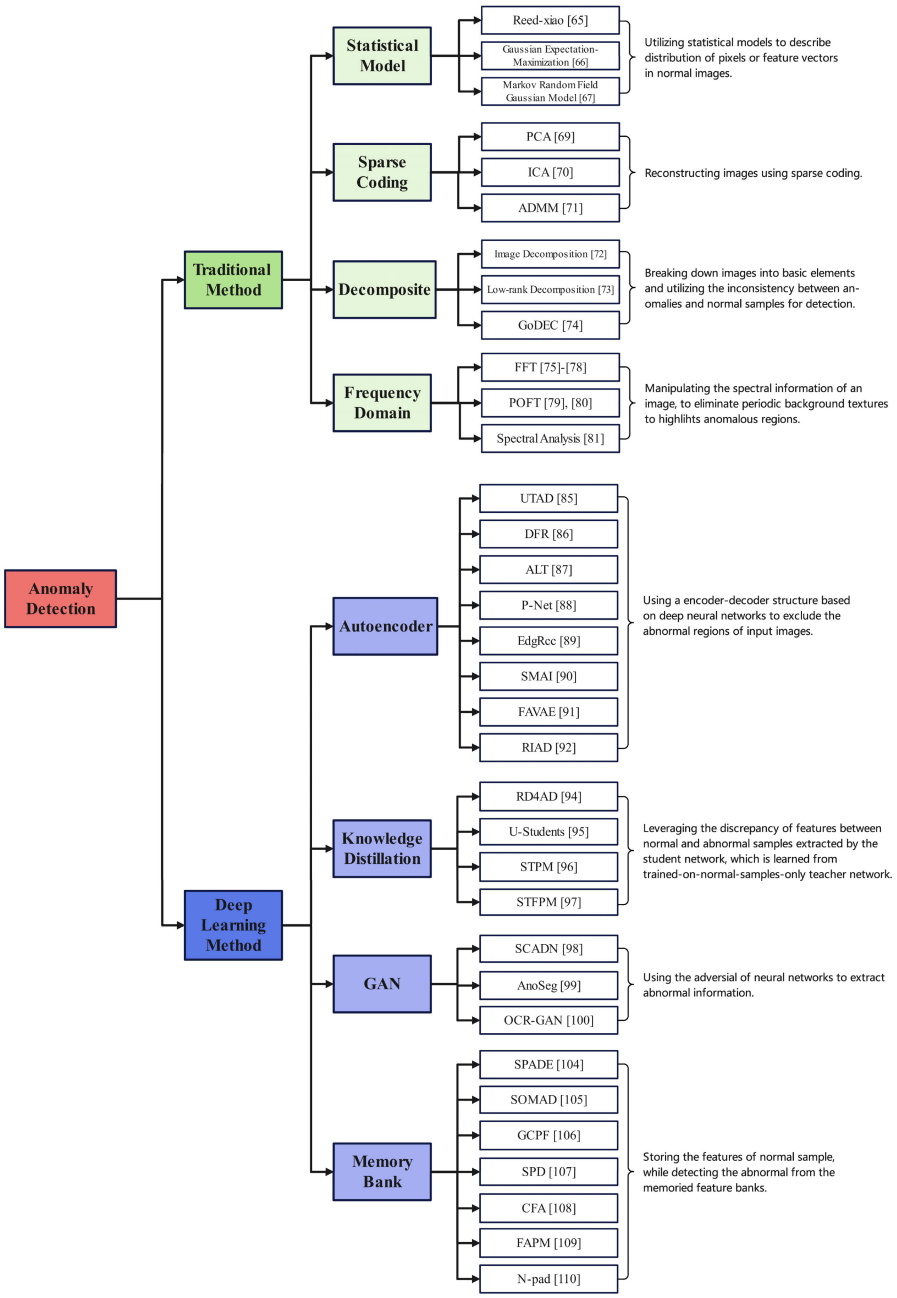


Fig. 3. Anomaly detection methods overview.

1) Statistical Model-Based Methods: Statistical model-based methods typically utilize statistical models to describe the distribution of pixels or feature vectors in normal images. Regions in images that deviate from this distribution are considered anomalies. These methods typically represent the normal parts of images using Gaussian distributions in the spatial or frequency domain, detecting abnormal regions based on outliers. Representative works include Reed–Xiao (RX) [65], Gaussian expectation–maximization [66], Markov random field Gaussian model [67], and so on.

The RX algorithm was selected as the representative method for detailed analyses. The RX algorithm is based on two hypotheses, H_0 and H_1. Assuming that the background and the anomaly target have the same covariance matrix $\mathbf{C}_b$, if the target is present, the mean of the background becomes the feature vector $\mathbf{s}$ of the target. H_0 represents the absence of a target, and at this time, the background follows an $N(0, \mathbf{C}_b)$ distribution; H_1 represents the presence of a target, and the background follows an $N(\mathbf{s}, \mathbf{C}_b)$ distribution.

For the input image that consists of P spectral bands, the background containing N pixels can be represented as a P^*N matrix $\mathbf{X}_b = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N]$, where $\mathbf{x}_i = [x_1, x_2, \ldots, x_P]^{\mathrm{T}}$. Under the H_0 hypothesis, the mean vector μ_b and covariance matrix $\mathbf{C}_b$ of the background are estimated as follows:

$$\mu_b = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i, \quad \mathbf{C}_b = \frac{1}{N}\mathbf{X}_b\mathbf{X}_b^{\mathrm{T}}. \tag{12}$$

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For each pixel $\mathbf{x}_i$ with P spectral bands, its RX statistic r is calculated as follows, which is equivalent to the Mahalanobis distance [68]:

$$r = (\mathbf{x}_i - \mu_b)^{\mathrm{T}} \mathbf{C}_b^{-1} (\mathbf{x}_i - \mu_b). \tag{13}$$

Given a threshold λ, if a pixel's RX statistic r is larger than the threshold, then the pixel is determined to be an anomalous one, which conforms more to the H_1 hypothesis (the target is present); otherwise, it is a background pixel under H_0 hypothesis (the target is absent).

2) Sparse Coding Reconstruction-Based Methods: Methods of this kind typically involve reconstructing images using sparse coding. In this process, a dictionary is learned to represent normal images. Subsequently, during the testing phase, anomaly detection is performed based on aspects such as reconstruction differences and sparsity. This includes methods such as principal component analysis (PCA) [69], independent component analysis (ICA) [70], and alternating direction method of multipliers (ADMMs) [71].

Analyzing with the classical PCA algorithm as a representative, suppose that there are m normal images, and each image is flattened into a vector $\mathbf{x} = [x_1, x_2, \ldots, x_n]^{\mathrm{T}}$ with n pixels; then, an $m^* n$ matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m]$ can be conducted.

For the matrix $\mathbf{X}$, the covariance matrix $\mathbf{C}$ can be computed first. The eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ and corresponding eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ of $\mathbf{C}$ are then computed. The top k principal eigenvectors are selected with the larger eigenvalues and are rearranged as the projection matrix $\mathbf{P} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_k]$, where k is far smaller than n.

In the testing phase, the flattened vector $\mathbf{x}$ of test image can be mapped to a k-dimensional vector $\mathbf{y}$ in the principal component space, which is calculated by $\mathbf{y} = \mathbf{P}^{\mathrm{T}}\mathbf{x}$. Then, the reconstruction vector $\mathbf{x}'$ is computed as follows:

$$\mathbf{x}' = \mathbf{P}\mathbf{y}. \tag{14}$$

For each pixel x_i of the test image, where $i = 1, 2, \ldots, n$, the PCA anomaly score can be computed in a typical approach, namely, weighted summation of all possible k values. Suppose that there are q possible k values in total, the anomaly score c is calculated using the following equations:

$$c = \sum_{p=1}^{q} |x_i - x_i'| \cdot w_p \tag{15}$$

$$w_p = \frac{\sum_{j=1}^{p} \lambda_j}{\sum_{j=1}^{n} \lambda_j} \tag{16}$$

where w_p is the weight of the pth possible k values, with p principal eigenvectors selected.

3) Image Decomposition-Based Methods: Methods based on image decomposition are mostly designed for the detection of small anomalous areas on surfaces with periodic textures. Since anomalous regions typically appear randomly and have weaker periodicity, this characteristic allows them to be distinguished from backgrounds with strong periodic textures. Classical methods include hyperspectral image decomposition [72], low-rank decomposition [73], GoDec [74], and so on.

DEC represents classical low-rank decomposition methods. For the input image, low-rank decomposition is employed to decompose the original test image into a low-rank matrix representing the background and a sparse matrix representing the anomalous regions. It is represented as follows:

$$\min_{\mathbf{L},\mathbf{S}} ||\mathbf{L}||_* + \lambda ||\mathbf{S}||_1, \quad \text{s.t. } \mathbf{F} = \mathbf{L} + \mathbf{S} \tag{17}$$

where $\mathbf{F}$ represents the original image matrix; $\mathbf{L}$ and $\mathbf{S}$ represent the low-rank matrix and sparse matrix, respectively; $||\cdot||_*$ represents the rank of matrix; $||\cdot||_1$ represents the L1 norm to approximately describe the sparsity of the matrix; and λ is the tuning hyperparameter.

4) Frequency-Domain-Based Methods: Frequency-domain-based methods mainly manipulate the spectral information of an image, attempting to eliminate periodic background textures to highlight anomalous regions. This method can be categorized into two domains: background spectrum elimination and phase-only Fourier transform. The former aims to eliminate periodic background spectral information to highlight anomalous regions, while the latter separates repetitive background and anomalous regions by extracting only the phase spectrum.

For the former, most existing works [75], [76], [77], [78] first utilize the Fourier transform to convert the original image into the frequency domain. After removing the spectral components corresponding to periodic background textures in the magnitude spectrum, the inverse Fourier transform is applied to obtain the spatial information of anomalous regions.

For the phase-only-based methods [79], [80], [81], the amplitude spectrum is discarded, with only the phase spectrum being subjected to inverse discrete cosine transform (IDCT), as shown in (18)–(20)

$$\boldsymbol{\mathcal{X}} = F(\mathbf{I}) \tag{18}$$

where (18) denotes the discrete cosine transform (DCT) processing, $F(*)$ is the Fourier transform, $\mathbf{I}$ refers to the original image, and $\boldsymbol{\mathcal{X}}$ is the image in the frequency domain with r rows and c columns

$$\boldsymbol{\mathcal{X}}'(u, v) = \frac{\boldsymbol{\mathcal{X}}(u, v)}{\mathbf{M}(u, v)} \quad \forall u \in [0, r) \ \forall v \in [0, c) \tag{19}$$

where (19) represents the extraction of the phase spectrum, $\boldsymbol{\mathcal{X}}(u, v)$ is the value at position (u, v) after the Fourier transform, $\mathbf{M}(u, v)$ represents the amplitude value at that point, and $\boldsymbol{\mathcal{X}}'(u, v)$ is the normalized result. This process can remove the majority of periodic textures from the original image, highlighting the anomalous regions [82]

$$\mathbf{I}' = F^{-1}(\boldsymbol{\mathcal{X}}') \tag{20}$$

where (20) is the IDCT process, $F^{-1}(*)$ represents the inverse Fourier transform, $\boldsymbol{\mathcal{X}}'$ is the normalized image in the frequency domain, and $\mathbf{I}'$ refers to the output image through phase-only-based methods.

B. Anomaly Detection Methods Based on Deep Learning

Deep learning-based methods do not rely on manually designed features; instead, they leverage DNNs to automatically learn extracting discriminative features, providing the

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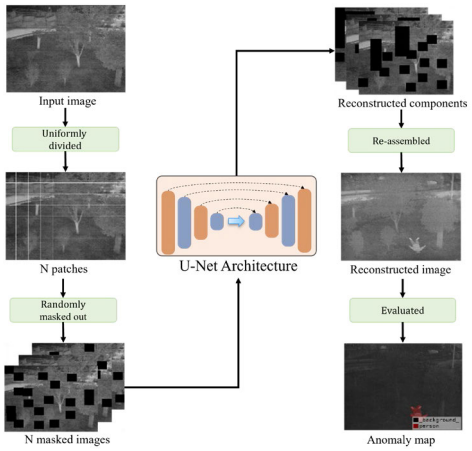


Fig. 4. Pipeline of RIAD. Initially, the input image is divided into n patches of equal size. These patches are then subjected to random masking, a process that is repeated n times to yield n masked images. It is imperative to ensure that the intersection of masked patches in each image is devoid of any content. Subsequently, the covered portions of the n masked images are reconstructed by a U-Net architecture model, and the reconstructed components are then assembled to obtain a complete reconstructed image. Finally, following evaluation of the results, an anomaly map is generated.

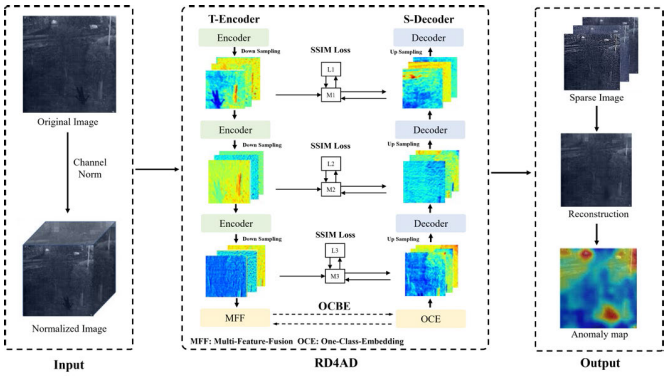


Fig. 5. Pipeline of RD4AD. The model itself consists of three components: a pretrained teacher encoder (T-Encoder), a trainable student decoder (D-Decoder), and a OCBE module. The OCBE module includes multifeature fusion (MFF) module and the one-class-embedding (OCE) module. Also, the SSIM loss in the illustration refers to the similarity loss.

algorithm with higher generalization capability. The existing methods can be broadly categorized into four types, namely, AE-based, knowledge distillation-based, generative adversarial network (GAN)-based, and memory bank-based approaches.

1) AE-Based Methods: These methods typically utilize the architecture of AE [83] or variational AE (VAE) [84] to extract features. For an input image $\mathbf{X}$, the process begins by obtaining the latent code $\mathbf{X}'$ of $\mathbf{X}$ through an encoder. Subsequently, $\mathbf{X}'$ is decoded by a decoder to produce $\mathbf{Y}$. Due to the anomaly-free nature of the training process for the encoder–decoder, it can only learn features from normal samples and cannot reconstruct anomalous regions. Therefore, the difference between $\mathbf{Y}$ and $\mathbf{X}$ can serve as an indicator for detecting anomalies. Representative works include: UTAD [85], DFR [86], ALT [87], P-Net [88], EdgRec [89], SMAI [90], FAVAE [90], and RIAD [92].

For detailed analysis, RIAD was selected, as shown in Fig. 4. The input image is uniformly divided into n patches of size k^*k. n patches are randomly masked out, and this process is repeated n times to obtain n masked images. It is ensured that the intersection of masked patches in each image is empty. During the model processing, the covered portions of the n masked images are reconstructed, and the reconstructed components are then assembled to obtain a complete reconstructed image.

2) Knowledge Distillation-Based Methods: Knowledge distillation aims to transfer the knowledge contained in the pretrained teacher model to the smaller student model, thereby achieving model compression [93]. In anomaly detection, knowledge distillation is applied to constrain the generalization capability of the encoder–decoder, disrupting the reconstruction of anomalous samples, thereby magnifying the distance between normal and anomalous samples in evaluation metrics [94], [95], [96], [97].

To replicate the experiments on the MRSI-NERD dataset, the current state-of-the-art reverse knowledge distillation method, namely, RD4AD [94], was selected. The process is illustrated in Fig. 5. For the input image, channel normalization was first performed and then fed into the model. The model consists of a fixed, extensively pretrained teacher encoder (T-Encoder), a trainable student decoder (D-Decoder), and a one-class bottleneck embedding (OCBE). The input image is encoded by T-Encoder, undergoes redundancy removal through the OCBE module, and is then decoded by S-Decoder with the expectation of restoring the normal image information. The similarity loss between encoder and decoder is calculated, and the optimization is performed by backpropagating gradients to update decoder and OCBE.

3) GAN-Based Methods: The application of GANs in image generation is widespread. Some efforts have been made to constrain the generator to learn features only from normal samples, preventing it from generalizing to anomalous samples. There are various works on GAN-based anomaly detection [98], [99], [100], and here, omni-frequency channel-selection representations (OCR)-GAN was selected as a representative example [100], as shown in Fig. 6.

For the generator, frequency decoupling is performed to decompose input images at different frequencies. This step is achieved through predesigned Gaussian blurring. The decoupled inputs are sequentially fed into generators corresponding to different frequencies through a channel selection module to reconstruct the respective images. The final output is obtained by accumulating these reconstructions.

For the discriminator, it takes the original image $\mathbf{X}_1$ and the reconstructed image $\mathbf{X}_2$ as input $\mathbf{X}_1$ serves as negative inputs, representing normal samples, while $\mathbf{X}_2$ and the augmented samples of $\mathbf{X}_1$ serve as positive inputs, representing abnormal samples. During the training phase, only $\mathbf{X}_1$ and its augmented version (referred to as CutPaste [101] and CutOut [102] in the original text) are used as positive and negative samples, respectively. This is done to learn the distribution of normal images and train the discriminator. During the testing phase, the discriminator produces binary classification labels (normal/abnormal), and the location of anomalies is identified through the reconstruction error.

4) Memory Bank-Based Methods: To address the issue where neural networks can reconstruct unseen anomalous regions, memory bank-based methods store features of nor-

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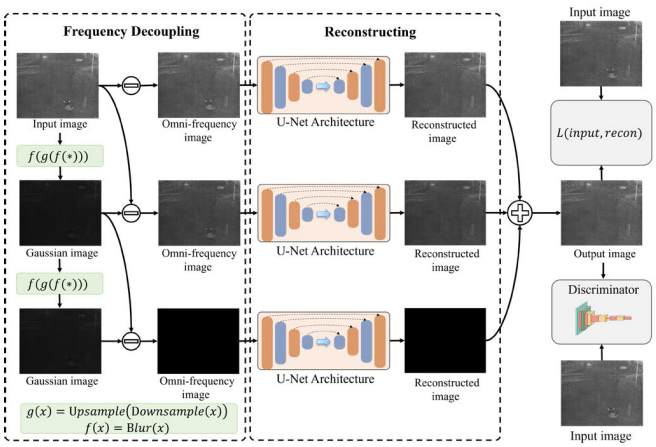


Fig. 6. Pipeline of OCR-GAN. OCR-GAN comprises two distinct steps: frequency decoupling and reconstruction. In the frequency decoupling step, the input image is decomposed at different frequencies through predesigned Gaussian blurring. In the reconstruction step, through a Channel Selection module, the decoupled inputs with different frequencies are then fed sequentially into a corresponding U-Net architecture generator. The generators are used to reconstruct the respective images and the final output is obtained by accumulating these reconstructions. In the training phase, the discriminator processes the original image and its augmented version to learn the distribution of normal images. In the testing phase, the discriminator produces binary classification labels (normal/abnormal), and the location of anomalies is identified through the reconstruction loss.

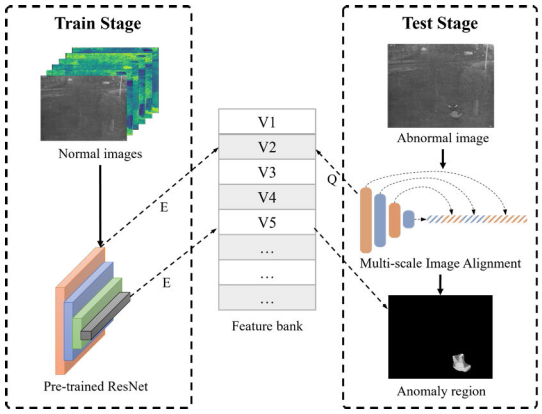


Fig. 7. Pipeline of SPADE. In the training stage, SPADE employs a pretrained ResNet to extract features from normal images. The features are subsequently stored in a feature bank. During the test stage, a multiscale image alignment is utilized for anomaly localization. This involves querying the features in the feature bank to generate an anomaly map. In the illustration, "E" denotes "extract features" and "Q" signifies "query."

mal samples as prototypes. During decoding, the decoder indexes the stored features for reconstruction, thereby limiting the CNN's generalization ability to unseen features. These fields include many classic works, such as PatchCore [103], SPADE [12], SOMAD [105], GCPF [106], SPD [107], CFA [108], FAPM [109], and N-pad [110].

Selecting SPADE as a general representative, as shown in Fig. 7, this method extends the KNN anomaly detection algorithm from image level to pixel level. During the training phase, a ResNet feature extractor pretrained on the ImageNet dataset is used to extract features from normal images, and a feature bank is maintained. During the inference phase, first, image-level KNN matching is performed to distinguish whether an image is normal or abnormal. For abnormal images, multiscale image alignment is introduced for pixel-level anomaly localization. For a normal sample that is misclassified as abnormal, the pixel-level localization should output a mask of all zeros.

V. Experiments and Analysis

A. Experimental Settings

1) Datasets: In this article, the dataset was organized following the format of MVTec, where each learning-based algorithm was trained on the corresponding target-free sample scene classification. There are a total of four different scenes for each modal, each containing 1–3 targets. To validate the effectiveness of the fusion method, the fused data were synthesized using the fusion algorithm proposed and conducted the same validation analysis. The results are presented in Section V-D.

2) Metrics: For the evaluation of fusion, EN, SD, SF, MI, SCD, and SSIM were chosen as comprehensive metrics. Suppose that the fused image is **F**, and the source images are **A** and **B**. Also, for an image **I** with size $M \times N$, assuming that there are L gray levels, μ_I presents the mean value of the image.

EN is a metric used to measure the amount of information in an image. The formula for EN can be presented as follows, where $p(i)$ represents the probability of a pixel with gray value i appearing:

$$\mathrm{EN} = -\sum_{i=1}^{L} p(i) \log_2 p(i). \tag{21}$$

SD reflects the dispersion degree of pixel gray values in an image relative to the mean value. The formula for SD can be expressed as follows:

$$\mathrm{SD} = \sqrt{\frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} (\mathbf{I}(i, j) - \mu_I)^2}. \tag{22}$$

SF refers to the rate of change of the greyscale of an image. In general, the greater the SF, the higher the quality of the fused image.

SF includes two further elements, namely, row frequency (RF) and column frequency (CF). These are described by (23), where RF and CF are calculated by (24) and (25), respectively,

$$\mathrm{SF} = \sqrt{\mathrm{RF}^2 + \mathrm{CF}^2} \tag{23}$$

$$\mathrm{RF} = \sqrt{\frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} (\mathbf{I}(i, j) - \mathbf{I}(i, j-1))^2} \tag{24}$$

$$\mathrm{CF} = \sqrt{\frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} (\mathbf{I}(i, j) - \mathbf{I}(i-1, j))^2}. \tag{25}$$

MI is used to measure the amount of information transferred from the source image to the fused image. The higher the MI of the fused image, the more information is transferred from the source to the fused.

The total MI can be presented in (26). For the source image **A**, the MI between the source **A** and the fused image **F** is described as (27), where h_A and h_F denote the edge histogram

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of them and $h_{A,F}$ denotes the joint histogram of the source and fused images

$$\mathrm{MI} = \mathrm{MI}_{A,F} + \mathrm{MI}_{B,F} \tag{26}$$

$$\mathrm{MI}(A, F) = \sum_i \sum_j h_{A,F}(i, j) \log_2 \frac{h_{A,F}(i, j)}{h_A(i) h_F(j)}. \tag{27}$$

SCD is a measure of the discrepancy between the fused image and the source image, which is used to evaluate the efficacy of the fusion algorithm. It can be observed that an elevated SCD value indicates a greater degree of information content present in the source image.

The formula for SCD can be described by (28), as shown at the bottom of the next page, where $D_{A,F}$ denotes the difference between the fused image $\mathbf{F}$ and the source image $\mathbf{A}$ (similarly for $D_{B,F}$). For any two images $\mathbf{X}$ and $\mathbf{Y}$, $r(X, Y)$ is presented in (29), as shown at the bottom of the next page.

SSIM is employed to quantify the information loss and distortion during the fusion process. A higher SSIM value indicates a greater structural similarity between the two images, implying a lower information loss and distortion during fusion.

The total SSIM can be presented in (30), and the SSIM between two images $\mathbf{X}$ and $\mathbf{Y}$ is described as (31). σ_X presents the SD of image X (similarly for σ_F), where $C_1 = (0.01 \times L)^2$ and $C_2 = (0.03 \times L)^2$

$$\mathrm{SSIM} = \mathrm{SSIM}_{A,F} + \mathrm{SSIM}_{B,F} \tag{30}$$

$$\mathrm{SSIM}_{X,F} = \frac{(2\mu_X\mu_F + C_1)(2\sigma_{XF} + C_1)}{(\mu_X^2 + \mu_F^2 + C_1)(\sigma_X^2 + \sigma_F^2 + C_2)}. \tag{31}$$

For the testing of anomaly detection, the settings of MVTec AD [111] were followed, adopting three popular metrics to get a comprehensive performance evaluation of all benchmarks, including traditional and deep learning-based methods, namely, FPR, TPR, and AUC. Negative denotes the background pixels in the image without target objects.

FPR in anomaly detection measures the proportion of normal data points that the model incorrectly identifies as anomalies, representing the rate of false alarms in a model.

It is described in the following equation, where false positive (FP) is the number of negative cases that are incorrectly predicted as positive and true negative (TN) is the number of negative cases that are correctly predicted as negative:

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}. \tag{32}$$

TPR is a crucial metric that measures the proportion of actual anomalies that the detection algorithm correctly identifies, reflecting the ability of a model to identify the positive instances accurately.

It is presented in (32), where true positive (TP) is the number of positive cases that are correctly predicted as positive and false negative (FN) is the number of positive cases that are incorrectly predicted as negative

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{33}$$

AUC provides a comprehensive evaluation of the performance of an anomaly detection method across different decision thresholds. The receiver operating characteristic (ROC) curve is plotted with TPR on the y-axis and FPR on the x-axis for various threshold settings. The AUC is the area under this ROC curve.

3) Tested Approaches: For image fusion, the proposed method was compared with a wavelet transform and the state-of-the-art deep learning-based methods, including U2Fusion [112], SDNet [113], TarDAL [114], and DeFusion [115]. For traditional anomaly detection algorithms, the following were tested: 1) RXD as a statistical model-based method; 2) PCA as a sparse coding reconstruction-based method; 3) DEC as an image decomposition-based method; and 4) POFT as a frequency-domain-based method. For deep learning-based anomaly detection algorithms, the following were tested: 1) RIAD as AE-based methods; 2) RD4AD as a knowledge distillation-based method; 3) OCR-GAN as a GAN-based method; and 4) SPADE as a memory bank-based method. All parameters were carefully tuned to achieve the best results.

4) Parameter Settings: For the wavelet transform, the wavelet basis function was the second-order Daubechies wavelet, with the decomposition level set to 1, where the fusion strategy was using the average method for low frequency and the maximum method for high frequency. For the deep learning-based fusion methods, the parameter settings were adjusted to be consistent with the references. For the fusion method proposed, the number of epochs for training was set to 150 with 60 and 90 epochs in the first and second stages, respectively. The batch size was set to 32. For the model hyperparameter settings, the number of GRL blocks in the multimodal feature encoder was 6, with eight attention heads and 64 channels. The channel of the Lite-transformer-based block was also 64 with eight attention heads. The configuration of the decoder is the same as that of the encoder. As for loss functions (6), $\alpha 1$–$\alpha 3$ were set to 0.3, 0.3, and 0.4, respectively. As for loss functions (7), μ was set to 0.5. As for loss functions in (9), λ_1–λ_5 were all set to 0.2. For the anomaly detection algorithms, the summary of the percentage of larger eigenvalues is over 99%, the tuning hyperparameter was set to 1 in DEC, and the hyperparameters in RIAD, RD4AD, OCR-GAN, and SPADE were consistent with the references.

B. Comparison of the Traditional Methods

As shown in Fig. 8, in the low-light modality, the traditional methods performed poorly in all four scenarios. The binary and heat maps of POFT and PCA were complex and lacked the ability to be recognized. The DEC resulted in almost blank binary maps, with fewer high values (i.e., hotter regions) being detected. However, RXD was effective in detecting anomalies, particularly in STD scenarios, and showed suboptimal performance in other scenes compared to the other methods.

As shown in Fig. 9, in the infrared modality, the four methods had a superior performance than the low-light modality in all four scenarios. It is obvious that the person was easier to recognize in the heat maps of DEC and POFT, but the binary maps of DEC were still empty. The PCA heat maps had less noise. In addition, the RXD exhibited superior performance, especially in the STD, FAN, and OAH scenes,

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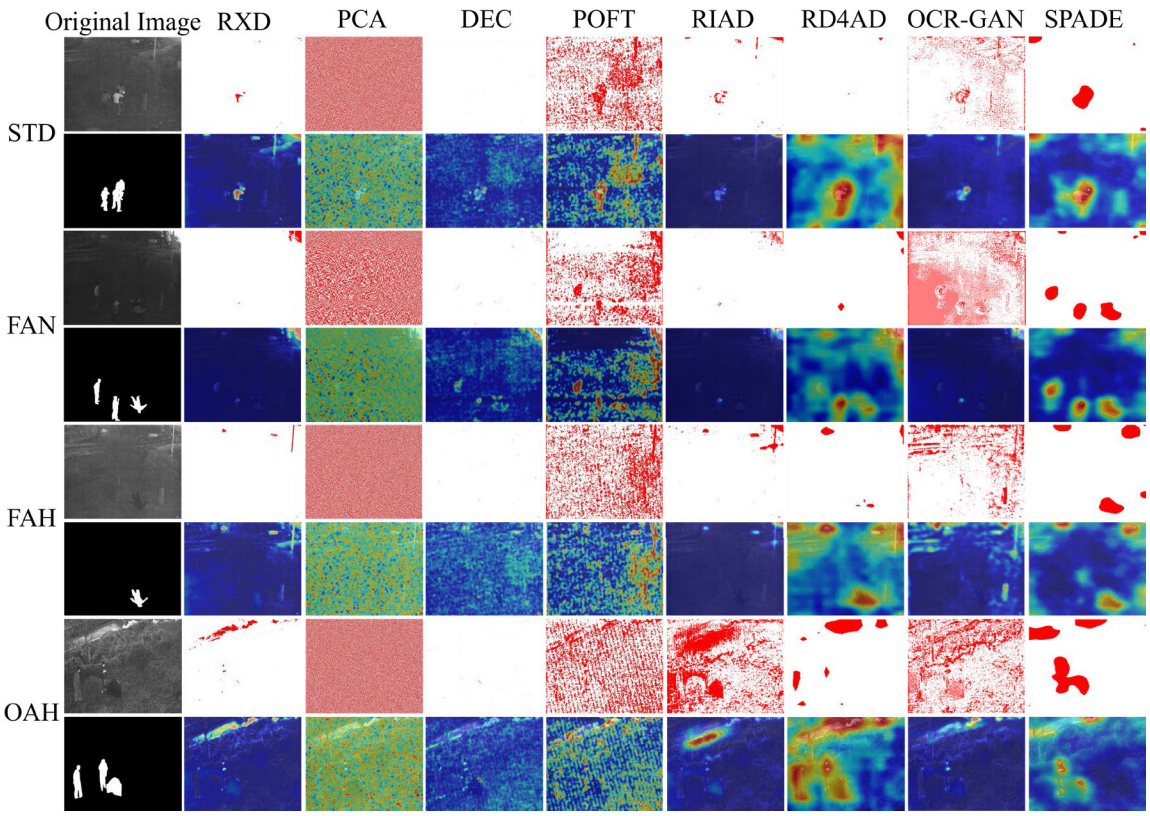


Fig. 8. Performances of the traditional and deep learning detection algorithms on the low-light dataset.

with obvious objects detected in both binary and heat maps, where high anomaly scores (i.e., the hotter regions) were more concentrated on the person to be rescued.

The results presented in Table I demonstrate that, among all algorithms, the RXD algorithm achieved the best performance. PCA and DEC struggled to cope with complex nighttime rescue anomaly detection, resulting in poorer performance. This may be PCA's inability to capture the complex non-linear relationships and DEC's difficulties due to its low-rank distribution assumptions. The POFT algorithm performed well in a few scenarios, as indicated by the AUC metric, where the frequency domain may contain more information and structure to detect. However, traditional detection algorithms generally face challenges of high FPRs, making it difficult for direct deployment.

C. Comparison of the Deep Learning-Based Methods

As shown in Fig. 8, in the low-light modality, the anomaly regions detected by the SPADE algorithm were able to encompass the person to be rescued, particularly in the STD, FAN, and FAH scenes. However, the binary maps of OCR-GAN were complex, with fewer high values (i.e., hotter regions) detected on the ground truth. The binary maps of RIAD and RD4AD did not cover the person, and instead, the heat maps of RD4AD covered the regions of the anomaly.

As shown in Fig. 9, the deep learning-based methods achieved better performance in the infrared modality compared to the traditional algorithms proposed above. Compared to the low-light modality, RIAD and OCR-GAN detected more edges of the ground truth in the binary maps. Specifically, SPADE performed poorly in the OAH scenario, detecting misplaced regions (i.e., background) in both binary and heat maps.

The results in Table I show that the SPADE algorithm based on memory banks outperformed others in most of the recognition tasks, followed by the state-of-the-art knowledge distillation method RD4AD. On the other hand, methods based on GAN, such as OCR-GAN, and those using E–D structures, such as RIAD, showed relatively poorer performance on this dataset.

D. Performance of Fusion-Based Detections

The fusion results of the low-light and the infrared images are shown in Table II. The experimental results demonstrate that the proposed method has excellent performance on almost

$$\mathrm{SCD} = r(A, D_{A,F}) + r(B, D_{B,F}) \tag{28}$$

$$r(X, Y) = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(\mathbf{X}(i,j)-\mu_X)(\mathbf{Y}(i,j)-\mu_Y)}{\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{N}(\mathbf{X}(i,j)-\mu_X)^2\sum_{i=1}^{M}\sum_{j=1}^{N}(\mathbf{Y}(i,j)-\mu_Y)^2}} \tag{29}$$

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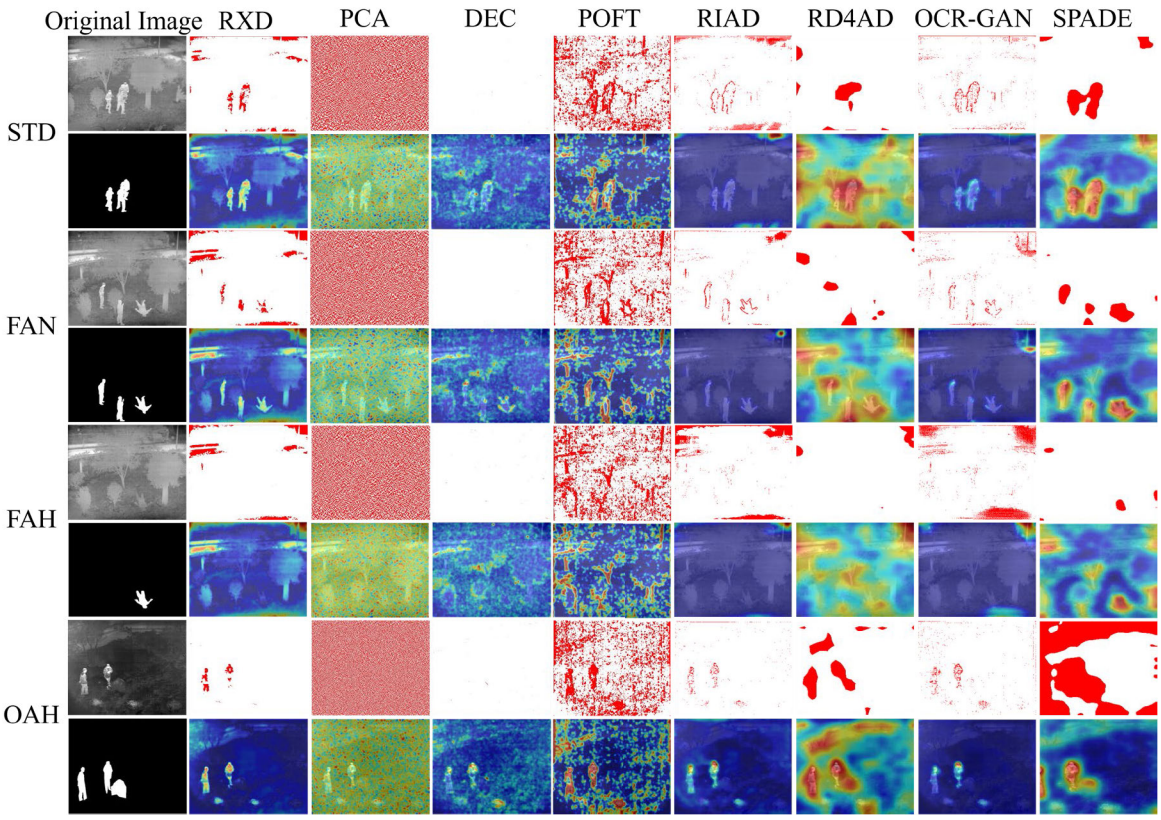


Fig. 9. Performances of the traditional and deep learning detection algorithms on the infrared dataset.

TABLE I
RESULTS OF TRADITIONAL METHODS AND DEEP LEARNING METHODS IN DETECTION

Metric	Modality	Scene	RXD	PCA	DEC	POFT	RIAD	RD4AD	OCR-GAN	SPADE
FPR↓	Low-Light	STD	0.981	**0.499**	0.990	0.630	0.813	0.881	0.871	0.742
	Low-Light	FAN	0.978	0.500	0.990	0.487	0.983	0.987	0.342	**0.150**
	Low-Light	FAH	1.000	0.497	1.000	0.569	1.000	0.701	1.000	**0.101**
	Low-Light	OAH	0.997	0.499	0.990	0.703	**0.492**	0.801	0.937	0.980
	Infrared	STD	0.895	0.895	1.000	0.630	0.652	0.881	0.920	**0.200**
	Infrared	FAN	0.610	0.517	1.000	0.388	0.849	0.824	0.840	**0.132**
	Infrared	FAH	1.000	0.498	1.000	0.598	0.997	0.999	0.990	**0.144**
	Infrared	OAH	0.980	**0.501**	0.990	0.700	0.970	0.815	0.959	0.845
TPR↑	Low-Light	STD	**0.997**	0.500	0.990	0.710	0.975	0.979	0.995	0.954
	Low-Light	FAN	**0.995**	0.499	0.990	0.759	0.990	0.571	0.624	0.891
	Low-Light	FAH	**0.995**	0.500	0.990	0.683	0.939	0.897	0.988	0.903
	Low-Light	OAH	0.983	0.499	0.990	0.661	0.789	0.421	0.929	**0.993**
	Infrared	STD	0.964	0.499	**0.990**	0.753	0.980	0.437	0.987	0.831
	Infrared	FAN	0.940	0.498	**0.990**	0.757	0.948	0.398	0.971	0.912
	Infrared	FAH	0.945	0.498	**0.990**	0.713	0.938	0.937	0.965	0.843
	Infrared	OAH	**0.990**	0.500	**0.990**	0.698	0.985	0.976	0.959	0.947
AUC↑	Low-Light	STD	0.454	0.500	0.493	0.541	0.660	0.643	**0.684**	0.680
	Low-Light	FAN	0.837	0.500	0.559	0.636	0.539	0.408	0.666	**0.931**
	Low-Light	FAH	0.826	0.502	0.491	0.557	0.526	0.744	0.368	**0.931**
	Low-Light	OAH	0.575	0.499	0.486	0.478	0.687	**0.867**	0.527	0.552
	Infrared	STD	0.562	0.499	0.509	0.562	0.699	**0.894**	0.569	0.715
	Infrared	FAN	0.829	0.489	0.555	0.683	0.640	**0.893**	0.673	**0.968**
	Infrared	FAH	0.285	0.503	0.490	0.557	0.442	0.477	0.399	**0.919**
	Infrared	OAH	0.447	0.499	0.504	0.499	0.483	**0.650**	0.486	0.607

all metrics, proving that our method is suitable for various kinds of scenarios. Specifically, the method proposed loses some MI (e.g., in STD and FAH scenarios) due to independence loss, but it still achieves better results compared to the baseline. Furthermore, the SD and SSIM of the proposed fusion method are less effective in the OAH scenario, which is related to the inherent contrast and environmental gradients in the OAH images.

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TABLE II
RESULTS OF FUSION METHODS

Scene	Method	EN↑	SD↑	SF↑	MI↑	SCD↓	SSIM↑
STD	Wavelet	6.52	26.46	5.1	2.3	1.98	0.87
	U2Fusion	6.59	26.74	7.4	**2.45**	1.91	0.94
	SDNet	6.61	27.05	8.42	1.99	1.89	1.13
	TarDAL	6.67	27.36	7.89	2.08	1.84	1.06
	DeFusion	6.64	27.89	7.15	2.12	1.79	1.19
	Ours	**6.7**	**27.95**	**8.83**	2.02	**1.77**	**1.22**
FAN	Wavelet	6.5	24.54	6.43	2.44	2.1	0.88
	U2Fusion	6.57	25.93	6.66	2.83	1.99	0.92
	SDNet	6.64	28.64	7.98	2.67	1.94	0.95
	TarDAL	6.73	27.12	8.12	2.56	1.88	0.94
	DeFusion	6.71	28.72	6.7	2.99	1.92	0.99
	Ours	**6.75**	**29.84**	**8.3**	**3.09**	**1.84**	**1.02**
FAH	Wavelet	6.43	27.93	8.13	1.98	2.05	0.97
	U2Fusion	6.50	28.1	8.45	2.04	1.95	1.09
	SDNet	6.46	28.67	8.71	1.41	1.96	1.22
	TarDAL	6.74	29.46	9.86	1.37	1.89	1.17
	DeFusion	6.72	29.63	8.42	**2.3**	1.83	1.25
	Ours	**6.81**	**29.94**	**9.89**	1.57	**1.77**	**1.27**
OAH	Wavelet	6.57	33.1	12.15	1.46	1.9	0.99
	U2Fusion	6.62	33.92	13.89	1.71	1.86	0.99
	SDNet	6.59	34.97	16.05	1.62	1.81	1.06
	TarDAL	6.67	34.6	15.51	1.5	1.75	1.03
	DeFusion	6.65	**35.43**	12.86	1.84	1.79	**1.12**
	Ours	**6.7**	31.22	**16.57**	**1.97**	**1.7**	1.07

TABLE III
RESULTS OF DETECTION ON FUSED IMAGES

Metric	Scene	RXD	RD4AD	SPADE
FPR↓	STD	0.686	0.850	**0.667**
	FAN	0.889	0.853	**0.143**
	FAH	1.00	0.879	**0.139**
	OAH	0.973	**0.943**	0.962
TPR↑	STD	**0.982**	0.978	0.962
	FAN	0.924	0.614	**0.944**
	FAH	**0.964**	0.944	0.906
	OAH	0.976	**0.979**	0.967
AUC↑	STD	0.906	**0.914**	0.740
	FAN	0.809	0.898	**0.974**
	FAH	0.318	0.731	**0.936**
	OAH	0.704	**0.869**	0.633

Boldface and underline show the best and second-best values, respectively. The SPADE outperforms in FAN and FAH scenarios, while the RXD and RD4AD algorithms, respectively, demonstrate superior performance in the FAN and FAH scenarios.

To further evaluate the performance of the fused images, downstream detection tasks were conducted on the fused dataset. RXD, which performed best among traditional algorithms, and RD4AD and SPADE, which exhibited excellent performance among deep learning algorithms, were selected. The results of the experiment are shown in Fig. 10 and Table III.

As illustrated in Fig. 10, the heat maps demonstrate that, in comparison with the single-modality-based, high anomaly scores (i.e., the hotter regions) are more concentrated on the person to be rescued, with diminished distribution over the background. Furthermore, in the binary maps, the anomaly regions cover the ground truth more comprehensively, demonstrating the improvements with fusion.

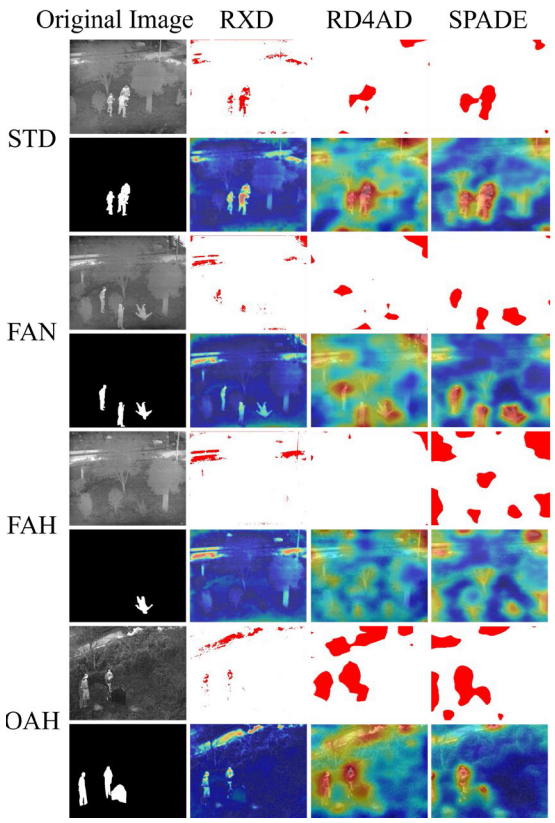


Fig. 10. Performances of the traditional and deep learning detection algorithms on the fused dataset. In comparison with the single modality, high anomaly scores (i.e., the hotter regions) in the heat maps are more concentrated on the person to be rescued, with diminished distribution over the background. Furthermore, in the binary maps, the anomaly regions cover the ground truth more comprehensively, demonstrating the improvements with fusion.

As demonstrated in Table III, the RXD and RD4AD algorithms, respectively, exhibit the most optimal performance under the STD and OAH scenarios on the fused dataset, whereas SPADE demonstrates superior performance under the FAN and FAH scenarios.

Boldface and underline show the best and second-best values, respectively. The experimental results demonstrate that the proposed method has excellent performance on almost all metrics, proving that our method is suitable for various kinds of scenarios.

Boldface and underline show the best and second-best values, respectively. The SPADE outperforms in FAN and FAH scenarios, while the RXD and RD4AD algorithms, respectively, demonstrate superior performance in the FAN and FAH scenarios.

To facilitate a comparison of the performance of RXD, RD4AD, and SPADE in different scenarios and multimodal datasets, a further illustration is provided in Fig. 11. The quantitative comparison reveals that the detection results of fusion outperform single modality in almost all scenarios. The SPADE based on fused images demonstrated improving AUC scores in all four scenarios, with higher TPR scores in STD, FAN, and OAH scenarios. Similarly, the RD4AD method,

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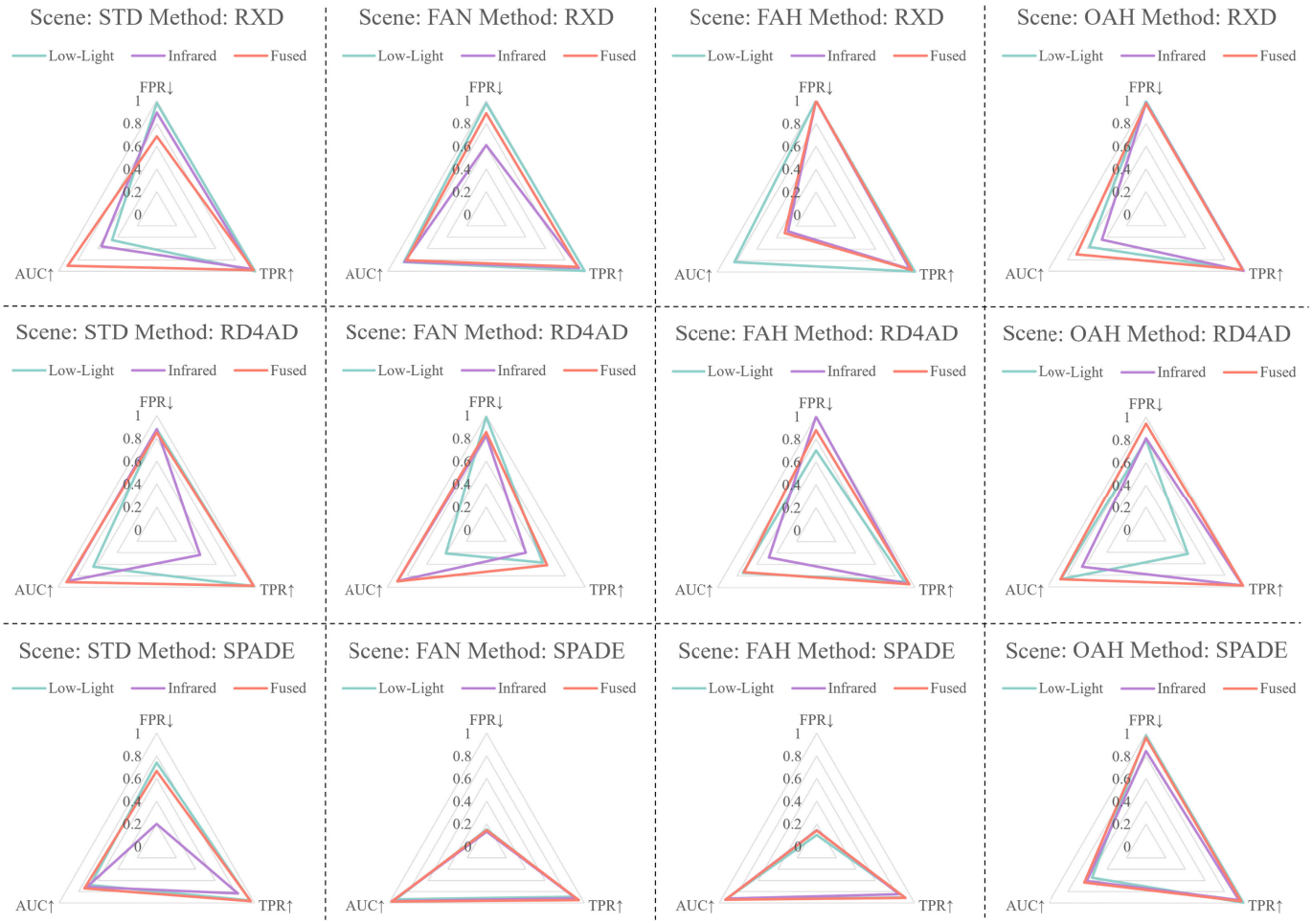


Fig. 11. Comparison of RXD, RD4AD, and SPADE in different scenes and multimodal datasets. The detection results of fusion have more superior AUC and TPR performance on almost all scenarios than single modality. The SPADE based on fused images demonstrated superior AUC scores in all four scenarios, with higher TPR scores in STD, FAN, and OAH scenarios. Similarly, the RD4AD method, based on the fusion, exhibited enhanced AUC and TPR values in nearly four scenarios (e.g., AUC values in the STD, FAN, and OAH). In the case of the RXD, the AUC of fusion surpassed the single modal in the STD and OAH scenarios, and TPR scores were superior in the STD and FAH. In addition, the performance of fused images in FPR exceeds that of the single-modal images only in certain instances (e.g., RXD in STD and OAH scenarios, and RD4AD in STD scenario).

based on the fusion, exhibited enhanced AUC and TPR values in nearly four scenarios (e.g., AUC values in the STD, FAN, and OAH). In the case of the RXD, the AUC of fusion surpassed the single-modal in the STD and OAH scenarios, and TPR scores were superior in the STD and FAH. However, the performance of fused images in FPR exceeds that of the single-modal images only in certain instances (e.g., RXD in STD and OAH scenarios, and RD4AD in STD scenario), where the misplaced regions are more localized in Fig. 10. Addressing this problem requires associating more features with contextual relationships, which is the limitation of our fusion approach for anomaly detection tasks in nighttime rescue missions.

VI. Summary and Discussion

In this article, a fair evaluation of mainstream unsupervised anomaly detection methods in the field of nighttime emergency rescue was conducted, and a fusion algorithm based on frequency-domain feature decomposition was proposed, which has been shown to enhance the performance of multiple detectors. To address the issue of insufficient data for target recognition and detection in nighttime emergency rescue situations, a well-annotated and augmented dataset (MRSI-NERD) was proposed. Considering the labor-intensive nature of constructing task-specific annotated datasets in real-world scenarios, which may not meet the timely requirements of emergency rescue, pixel-wise anomaly detection algorithms were researched. These algorithms treat a small number of annotated samples as anomaly information and extensively extract meaningful features from normal images. Thanks to the powerful feature extraction capabilities of DNNs, deep learning-based detection methods go beyond traditional algorithms. However, traditional algorithms based on RX also demonstrate decent performance in complex detection tasks. Among the deep learning methods, approaches based on reconstruction and memory bank have achieved superior performance. To further enhance the performance of unsupervised anomaly detection, a deep multimodal fusion algorithm based on frequency-domain feature decomposition was further proposed. Experimental results demonstrate that this fusion strategy effectively improves the performance of the detectors.

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In addition, due to the complexity of natural backgrounds, it has been observed that some anomaly detection methods struggle to meet the precision requirements for nighttime emergency rescue environments. Therefore, it is hoped that future research directions can focus on the following aspects.

1) Developing Unsupervised Domain Adaptation Methods: Anomaly detection methods rely on comprehensive normal datasets that are thoroughly annotated and representative of the operational environment. However, collecting such datasets can be challenging in nighttime emergency rescue scenarios due to the variability and unpredictability inherent to these situations. Unsupervised domain adaptation methods can effectively leverage data from both domains to learn an efficient anomaly detection model for the target domain [116]. Therefore, more unsupervised domain adaptation methods should be developed to promote the detection performance.

2) Developing Active Learning-Based Methods: The current process of labeling data is typically time-consuming and, as a consequence, a hindrance to the adoption of machine learning methods for the automated anomaly detection [117]. Employing expert feedback, active learning is an effective tool for building anomaly detection models. The model queries samples that need to be labeled by experts and then retrains the model using the samples. This approach can reduce the burden of acquiring labeled datasets while improving the anomaly detection performance [118]. Therefore, active learning-based low-light and infrared remote sensing images anomaly detection methods should be further developed.

3) Constructing Different Operators Based on Low-Light and Infrared Data: Nighttime operations are typified by low-light conditions, which can significantly restrict the visibility and efficacy of conventional imaging systems. While infrared data offer distinct advantages in low-light settings, it necessitates the utilization of specialized processing techniques. The characteristics of low-light and infrared data can be leveraged to construct operators that make full use of prior knowledge [119]. Thus, how best to construct the different operators based on low-light and infrared data, preserving texture details, performance, and operational efficiency, is an interesting topic.

4) Coupling Physical Mechanisms and Data-Driven Methods in Low-Light and Infrared Multimodal Detection System for Nighttime Rescue Mission: Due to the inability of capturing physical mechanisms present in the images, data-driven deep learning algorithms to image data frequently result in suboptimal performance in complex scenarios [120]. Some traditional anomaly detection algorithms, designed based on statistical or physical features of the image, such as RXD, demonstrate performance on par with some deep learning algorithms on the fused dataset. This suggests the potential to enhance the efficacy of low-light and infrared multimodal detection systems by coupling data-driven methods with physical mechanisms. Consequently, the combination of deep learning anomaly detection methods with physical mechanisms in low-light and infrared multimodal data could be promising for further investigation.

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Yuting Wan (Member, IEEE) received the B.S. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2017, the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2022.

He is currently a Post-Doctoral Researcher with Wuhan University. His major research interests include real-time remote sensing image processing, evolutionary computation, neural architecture search, and disaster emergency response.



Haoyu Yao is currently pursuing the bachelor's degree with the Remote Sensing and Information Engineering School, Wuhan University, Wuhan, China.

His major research interests include remote sensing, computer vision, multimodal learning, and embodied intelligence.



Jiawei Liu is currently pursuing the bachelor's degree with the School of Cyber Science and Engineering, Wuhan University, Wuhan, China.

His major research interests lie in remote sensing and artificial intelligence to software engineering.



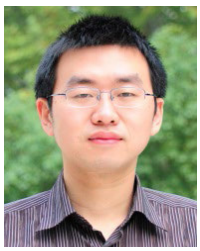
Chen Sun (Graduate Student Member, IEEE) received the B.S. and M.S. degrees from Wuhan University, Wuhan, China, in 2021 and 2023, respectively, where he is currently pursuing the Ph.D. degree in communication and information system with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS).

His research interests include multimodal object tracking and edge computing.



Ailong Ma (Senior Member, IEEE) received the B.S. degree from China University of Petroleum, Qingdao, China, in 2010, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

He is currently working as an Associate Professor at Wuhan University. He has authored or co-authored more than 30 research articles in *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION. His major research interests include multimodal remote sensing image processing, pattern recognition, and disaster damage assessment.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) research group. He has published more than 150 research papers in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He is currently serving as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and *International Journal of Remote Sensing*.

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